MATE-CLIP: Multimodal Adaptation and Temporal Enhancement for Action Recognition

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Abstract

Adapting a frozen CLIP to video action recognition is limited by fixed generic text templates that fail to capture the true action semantics, a frame-wise image encoder that cannot perceive how an action evolves, and a systematic misalignment between text and video features. We propose ***MATE-CLIP*** *(*Multimodal Adaptation and Temporal Enhancement CLIP*), a parameter-efficient framework that adapts and enhances both modalities and their interaction on a frozen backbone. On the text side, a static semantic branch is paired with a learnable branch refined by* AMTF*, a multi-strategy text-fusion module that yields action-biased prototypes. On the video side,* MGTA *selects action-critical regions and models them with a dual-branch design that combines dense local temporal modeling with sparse motion-correspondence attention. A lightweight* BFEM *then enhances each modality with the other under zero-initialized gates. All modules are identity-start, so the network coincides with the frozen baseline at initialization while adding few parameters. On UCF101 and HMDB-51, MATE-CLIP outperforms strong CLIP-based baselines, generalizes across the two datasets, and its modules contribute complementary gains.*

1. Introduction

Video action recognition seeks to identify human actions in videos. Contrastive language-image pre-training (CLIP) [19] has become a strong foundation for video understanding, and a line of work [17, 23, 24, 27] adapts frozen CLIP models to video action recognition with lightweight temporal modeling modules and learnable prompts, achieving strong performance without retraining the large backbone. Recent efforts extend this recipe to open-vocabulary action recognition [9] and to stronger video-text encoders [26, 31]. Despite this progress, the frozen CLIP text–video space still suffers from three under-explored semantic gaps that jointly limit recognition accuracy.

(i) Fixed generic templates fail to capture the true action semantics. A common recipe encodes each class with a fixed, handcrafted template such as "a video of a person {*action* class}", which is passed through the frozen CLIP text encoder to obtain a static class representation. This template is *generic*: it imposes the same rigid sentence structure on every category and does not adapt to the semantics of the specific action, so it can only express "which object or activity is present" rather than *how* the action is performed. Actions with similar wording but different motion—e.g., swinging a golf club versus carrying a golf club—are therefore pushed toward nearly identical text directions, because the template never describes the motion, the pose change, or the spatial relation that distinguishes them. The resulting text prototype is semantically shallow and static, which weakens inter-class separation whenever classes are described by the same template and share words or scenes.

(ii) Video-side temporal-dynamics deficiency and discriminative dilution. The frozen CLIP image encoder is inherently *frame-wise*: it encodes each frame independently and has no cross-frame receptive field, so it cannot perceive how an action evolves over time, and the onset, acceleration, direction change, and pose transition of a motion are all lost. Existing temporal adapters either aggregate all spatial-temporal tokens uniformly [27] or model the full token stream [13, 14], treating static background and action-critical moving regions equally. This uniform modeling is wasteful and, more importantly, *dilutes* the temporal dynamics and the discriminative signal of the critical regions: when every patch receives equal weight, the subtle movement of a hand or a body part is averaged away by static scenery.

(iii) Cross-modal directional misalignment. Even when both the text and video sides are enriched, the text class directions and the video class centers in the frozen CLIP space are systematically misaligned, because they are induced by distinct modalities, supervision signals, and temporal aggregations. Direct cosine classification under

this misalignment produces insufficient margins and unstable decisions.

We argue that these three gaps form an *information pipeline*: text prototypes serve as the *anchors*, video features serve as the *queries*, and the final classifier measures their *similarity*. Improving any single stage leaves the others untouched. For example, a richer text anchor is useless if the video query is diluted by background tokens, and even two well-formed representations fail if the directions they live in are systematically skewed relative to each other. Concretely, consider a golf swing. The `[EOT]`-only text prototype collapses the action description into a single vector, the uniformly pooled video feature averages the fast club motion into the static course scene, and the residual cross-modal shift further reduces the margin between swing variants. Each failure is attributable to a distinct stage of the pipeline, so a method that optimizes only one stage inevitably sacrifices recognition accuracy. This motivates a unified framework that strengthens all three stages in a complementary way rather than optimizing each module in isolation; it also suggests that the three modules should be *orthogonal* in design, each addressing its own stage, so that their gains add up.

In this paper, we propose **MATE-CLIP** (*Multimodal Adaptation and Temporal Enhancement CLIP*), a parameter-efficient framework that adapts a frozen CLIP backbone to video action recognition. Each of the three pipeline stages is addressed by one lightweight, plug-and-play module trained end-to-end on the *frozen* backbone: text adaptation (AMTF), video temporal modeling (MGTA), and cross-modal alignment (BFEM).

- **Adaptive Multi-Strategy Text Fusion** (AMTF) is the text-side module. It aggregates the `[EOT]`, masked-mean, and masked-attention views with learnable weights, yielding an action-biased and semantically complete prototype. AMTF runs on a learnable branch that is paired with a static semantic branch caching the frozen class semantics, so the final prototype retains the reliable frozen semantics while being biased toward the action.
- **Motion-Guided Temporal Adapter** (MGTA) enhances the video side. It injects temporal-dynamics modeling into the frozen image encoder, which otherwise has no cross-frame receptive field. MGTA first selects discriminative regions by fusing class-token attention, *motion intensity*, and a learnable saliency score. It then models them with a dual-branch design: a full-coverage Conv3d branch captures dense local temporal patterns, while an *asynchronous self-attention* branch treats motion as an *implicit correspondence*–each selected token predicts a per-token 3D offset to where the represented part has moved and reads features at that predicted location. This sparse motion-tracking cross-frame association captures how an action unfolds while concentrating computation on the action-critical regions.
- **Bidirectional Feature Enhancement Module** (BFEM) enhances each modality with the other before classification. Driven by a soft class assignment, it uses the video feature to enhance the text directions and uses the text prototypes to enhance the video feature, in both directions under bounded, zero-initialized gates that start from identity and open gradually during training. It therefore enhances without disturbing the already-converged adapters and needs no auxiliary alignment loss.

All three modules are zero-initialized or identity-start, so the frozen CLIP backbone is never modified and each module can be ablated independently.

Our main contributions are summarized as follows:

- We identify three complementary semantic gaps that jointly limit frozen CLIP video action recognition: fixed generic templates that fail to capture the true action semantics, video-side temporal-dynamics deficiency with discriminative dilution, and cross-modal directional misalignment.
- We propose MATE-CLIP, a parameter-efficient framework for multimodal adaptation and temporal enhancement, with three plug-and-play modules (AMTF, MGTA, BFEM) that go beyond the fixed generic text template to capture action-specific semantics, inject temporal dynamics into the frozen image encoder through motion-guided selection and implicit motion-correspondence attention, and explicitly align the two modalities, all on a frozen CLIP backbone.
- We show that the frozen-CLIP adaptation generalizes across datasets and that the three modules compose orthogonally, via systematic ablations on UCF101 and HMDB-51.

Experiments on UCF101 [20] and HMDB-51 [10] show that MATE-CLIP consistently outperforms strong CLIP-based baselines on both benchmarks and generalizes across the two datasets.

2. Related Work

2.1. CLIP for Video Action Recognition

CLIP [19] aligns image and text in a shared embedding space and has been widely adapted to video action recognition. ActionCLIP [23] applies CLIP to video with novel data augmentation and prompt ensembles, establishing a parameter-efficient recipe that keeps the backbone frozen. X-CLIP [17] introduces attention-based frame aggregation and video-specific prompts, and achieves strong results on general video recognition. Vita-CLIP [27] combines spatial-temporal modules with multimodal prompting and text-adapter alignment, and is among the strongest frozen CLIP baselines. M2-CLIP [24] further couples multimodal prompting with a multi-task objective to adapt

CLIP to video in a parameter-efficient way, and VLPA-CLIP [25] adapts CLIP with video-language prompting for efficient recognition, while FROSTER [9] shows that a frozen CLIP can serve as a strong teacher for open-vocabulary action recognition. More recent work strengthens the base representation itself, through stronger video-text encoders [26, 31] and better CLIP alignment [30], complementing adapter-based adaptation. Despite these advances, the above methods largely inherit two CLIP conventions: they use only the `[EOT]` token as the class representation, and they aggregate video tokens uniformly across space and time. As we analyze in Section 1, both conventions leave the three semantic gaps we target unaddressed, and simply strengthening the video side (as in X-CLIP or Vita-CLIP) does not fix the text-side or the cross-modal gap. At a higher level, these methods can be organized by which stage of the text–video pipeline they strengthen: prompt-engineering methods [17, 23, 25] enrich the video side with learnable prompts and aggregation, text-adapter methods [27] align video features to text, and retrieval-oriented methods [8, 28] refine the shared space. None of them strengthens all three stages simultaneously: the text prototype remains a single token, the video aggregation remains uniform, and the alignment remains directional. This systematic gap is precisely what MATE-CLIP addresses, through parameter-efficient multimodal adaptation and temporal enhancement with three complementary modules.

2.2. Video Temporal Modeling

Video temporal modeling has evolved from 3D convolutions [21, 22] and efficient shift-based designs [15] to two-pathway architectures [6] and attention-based transformers [1, 5, 14, 16]. These architectures are trained from scratch on large video corpora, which is costly and still underperforms image-pretrained models on small action datasets. A recent line of work instead adapts frozen image encoders to video with lightweight modules, such as ST-Adapter [18], ViT-Adapter [3], and AIM [29], drawing on the parameter-efficient transfer recipe of LoRA [7]. Parameter-efficient designs for CLIP video recognition [24, 25] and stronger video-text encoders [26, 31] continue this trend, yet most still model the full token stream and rarely exploit explicit motion correspondence across frames. However, most adapters still model the full token stream uniformly, so computation is spent on static background as well as on action-critical regions. Sparse token-selection methods [2, 13] reduce this redundancy by focusing on informative tokens. In particular, RAP [2] selects sparse discriminative patches with a correlated adapter and serves as a strong sparse baseline, but its selection is driven by text-video correlation and does not explicitly exploit motion across frames. Our MGTA module instead combines *discriminative region selection guided by motion* with *implicit motion-correspondence attention*. In the spirit of deformable attention [4], keys and values are sampled from query-dependent offset positions; crucially, however, MGTA predicts a per-token 3D offset that acts on the true spatial–temporal grid, so the offset encodes a *local* correspondence—"this part moves from here to there"—rather than a global uniform shift, which distinguishes it from shift-based temporal models [15]. The sparse branch thereby *tracks how action-critical regions move over time* and reads the features at their predicted new locations, acting as a lightweight, learnable motion model on top of the frozen encoder. Unlike token-pruning methods that remove tokens permanently, MGTA performs per-frame selection, so a moving part is re-located in each frame rather than dropped after its first appearance. And unlike dense 3D convolutions [6, 22], which apply the same kernel everywhere, MGTA spends its capacity only on the regions that actually change, which is both more accurate for subtle motion and cheaper at inference. Compared with the sparse baseline RAP [2], whose selection is driven by text–video correlation rather than by the dynamics of the video itself, MGTA makes the selection *motion-aware* and couples it with an explicit motion-correspondence branch, which is what allows it to capture how an action changes rather than only where it is.

2.3. Cross-Modal Alignment

Aligning vision and language features is central to CLIP and its successors [11, 12], which bootstrap image-text alignment with auxiliary objectives and frozen encoders. In video recognition, text-guided alignment [17, 27] aligns video features to text prototypes, and video-text retrieval methods [8, 28] refine the shared space with temporal and cross-modal constraints. More recent video-text encoders [26, 31] and CLIP alignment objectives [30] strengthen this alignment at the representation level, yet in closed-set classification the text directions remain frozen and are rarely corrected toward the actual video class centers. A key challenge is that direct alignment can disturb already-converged features. Gradient-based alignment toward text directions tends to pull video features away from their natural cluster structure, and the reverse happens when video prototypes are pushed into the text directions. Prior alignment methods typically train a projection from video to text or refine the text encoder itself. Both choices modify a whole stream of representations and therefore risk unlearning the discriminative structure that the frozen space already encodes. Our BFEM module instead enhances each modality with the other, using bounded, zero-initialized gates (identity start) and *soft feature enhancement* in both directions: the video enhances each text direction with a soft weight, and the text prototype enhances the video feature, both as convex combinations, while keeping the frozen

CLIP backbone intact. Because the gates open gradually during training, the enhancement starts from the already-converged baseline and only refines the features, rather than re-training the frozen features from scratch. Unlike projection-based alignment or hard-additive reconstruction, every enhanced feature stays on the segment between the original feature and its modal counterpart, so the enhancement can never drag a representation outside the shared embedding ball. This distinguishes BFEM from methods that map the video into the text side without ever refining the text directions.

3. Method

3.1. Overview

Problem setting. We study supervised video action recognition with a frozen CLIP backbone [19]. A video clip is sampled into T frames, and the frozen image encoder maps each frame into N patch tokens, yielding spatial-temporal tokens $F \in \mathbb{R}^{T\times N\times d}$ with embedding dimension d. A set of class prompts (one per category) is encoded by the frozen text encoder into token sequences, from which we obtain text prototypes $Q \in \mathbb{R}^{C\times d}$ for C classes. The goal is to map the video tokens F to a video feature $v \in \mathbb{R}^d$ and to classify v against Q with a cosine classifier, while keeping both CLIP encoders frozen. For a class index i, the classification logit is

$$\ell_i = \cos(v, Q_i) = \frac{v^\top Q_i}{\|v\|\,\|Q_i\|}, \tag{1}$$

and the plain frozen-CLIP baseline computes Q_i from the single `[EOT]` token of the i-th prompt and obtains v by uniformly pooling all frame tokens. Table 1 summarizes the notation used throughout this section.

Framework. Following Vita-CLIP [27], we adopt a frozen CLIP backbone as both the video and text encoders. MATE-CLIP augments this pipeline with three lightweight modules (Figure 1), each addressing one of the gaps identified in Section 1. On the text side, a static semantic branch is adaptively fused with a learnable branch refined by AMTF (Adaptive Multi-Strategy Text Fusion), producing class anchors that capture the true action semantics beyond what the fixed generic template provides. MGTA injects temporal-dynamics modeling into the frozen image encoder and concentrates video modeling on discriminative regions, producing a video feature v that captures how an action unfolds over time. BFEM bidirectionally enhances v and Q with each other before the final cosine classifier, so that each modality benefits from the other's discriminative information before classification. The three modules are complementary by construction: they act on the text anchors, the video queries, and the decision metric respectively, and each is zero-initialized or identity-start so that the network coincides with the plain baseline at initialization and the CLIP backbone is never modified. All three run in a single forward pass of the frozen backbone without iterative refinement.

Symbol	Meaning
T, N, d	#frames, #patches, embedding dim.
$F \in \mathbb{R}^{T\times N\times d}$	spatial-temporal patch tokens
$Q \in \mathbb{R}^{C\times d}$	text prototypes of C classes
$v, \hat{v}$	video feature and BFEM-enhanced feature
$\mathcal{S}$	selected discriminative region
g_t, g_v	zero-initialized enhancement gates in BFEM
$F^{\text{fix}}, q^{\text{learn}}$	fixed-branch and learnable-branch text features
$\lambda_e, \lambda_m, \lambda_a$	Text-MSTFF view fusion weights

Table 1. Notation used in the method.

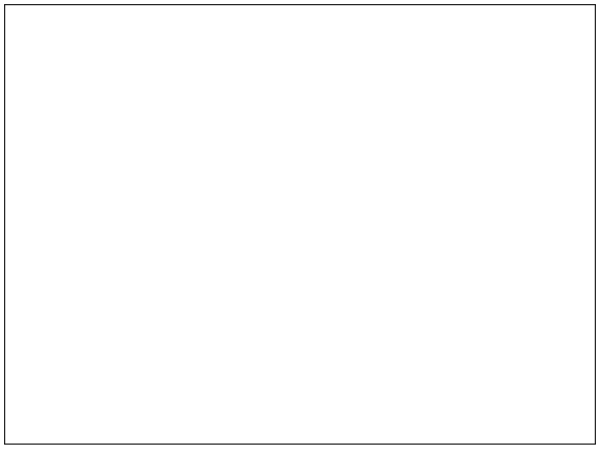


Figure 1. Overview of MATE-CLIP. The frozen CLIP backbone is shared by the text and video encoders. On the text side, a static semantic branch (frozen semantic bank) is combined with a learnable branch enhanced by AMTF (multi-strategy text fusion over the `[EOT]`, masked-mean, and masked-attention views), and the two branches are fused adaptively into action-biased prototypes. On the video side, MGTA selects discriminative spatial-temporal regions and performs dual-branch temporal modeling (dense Conv3d + sparse asynchronous attention); a lightweight frame-aggregation mixer (Mixer) then applies a depthwise temporal convolution to the frame features before the temporal mean. Finally, BFEM bidirectionally enhances the video feature and the text prototypes with each other under zero-initialized gates. The CLIP backbone remains frozen throughout.

3.2. AMTF: Adaptive Multi-Strategy Text Fusion

The text side of MATE-CLIP combines a *static semantic branch* with a *learnable branch* refined by the **AMTF** module. As discussed in Section 1, a fixed generic template such as "a video of a person {*action* class}" cannot describe *how* an action is performed, so actions that differ mainly in motion or pose collapse onto nearly identical text directions. The static branch preserves this frozen CLIP semantic as a

stable anchor, while the learnable branch with AMTF goes beyond the generic template to capture action-specific semantics; the two are fused adaptively at the branch level.

3.2.1. Static semantic branch

For each class i, the fixed template "a video of a person c_i" is encoded by the *frozen* CLIP text encoder, computed once, and cached as a buffer that is never trained. This yields a fixed text feature $F_i^{\text{fix}} \in \mathbb{R}^d$, L2-normalized, that acts as a *static semantic bank* anchoring the final prototype to the original CLIP semantics.

3.2.2. Learnable branch with AMTF

The learnable branch prepends a set of learnable context vectors (Text-Prompt-Learner) to the class tokens, encodes them with the same frozen encoder, and refines the sequence with the **AMTF** module (*Adaptive Multi-Strategy Text Fusion*, corresponding to the `Text_MSTFF` block). AMTF aggregates the token sequence $T = [t_1, \dots, t_L]$ with three complementary views, each capturing a distinct role: the *EOT view* $q_{\text{eot}} = t_L$ preserves the original CLIP semantic and guarantees that the frozen semantic information is never lost; the *masked-mean view*

$$q_{\text{mean}} = \frac{1}{\sum_l m_l} \sum_{l=1}^{L} m_l \, t_l, \tag{2}$$

summarizes the overall context of the whole sequence, with padding tokens excluded by the mask $m_l \in \{0, 1\}$; and the *masked-attention view*

$$q_{\text{attn}} = \sum_{l=1}^{L} \alpha_l \, t_l, \qquad \alpha = \text{softmax}(\psi(T) + (1 - \mathbf{m}) \cdot (-\infty)), \tag{3}$$

re-weights the tokens by a learned importance estimator ψ (a small MLP), amplifying the tokens most relevant to the task. Each view is produced by its own linear projection, and the three views are combined by *dynamic weighting* with learnable fusion weights:

$$q^{\text{amtf}} = \lambda_{\text{eot}} q_{\text{eot}} + \lambda_{\text{mean}} q_{\text{mean}} + \lambda_{\text{attn}} q_{\text{attn}}, \tag{4}$$

where λ is a softmax over learnable logits, initialized to the prior $1 : 0.5 : 0.5$ and bounded by per-view soft caps, so that the model adaptively balances the reliable frozen semantic (`[EOT]`), the overall context (masked-mean), and the task-relevant emphasis (masked-attention) without letting any single view dominate. The fused view is then combined with the branch's `[EOT]` feature in a residual way through a zero-initialized gate, $q_i^{\text{learn}} = \text{normalize}\big(t_L + \tanh(g)(q^{\text{amtf}} - t_L)\big)$, so the learnable branch starts from the frozen EOT semantic and opens the enhancement gradually.

3.2.3. Adaptive two-branch fusion

The two branch features are combined by a branch-level adaptive fusion with its own learnable weights:

$$Q_i = \text{normalize}\big(w_i^{\text{fix}} F_i^{\text{fix}} + w_i^{\text{learn}} q_i^{\text{learn}}\big), \tag{5}$$

where $(w_i^{\text{fix}}, w_i^{\text{learn}})$ is the softmax of a pair of logits $(\text{logit}^{\text{fix}}, \text{logit}^{\text{learn}})$ initialized to $(3.0, 0.0)$, i.e. approximately $(0.95, 0.05)$ after softmax. The prototype therefore starts dominated by the static semantic bank; as training proceeds, the learnable logit rises if the action-biased features help discrimination, adaptively re-balancing the trust between the reliable static anchor and the learned dynamic cues. Because both branches are identity-start, the text side is a parameter-efficient, plug-in design that never disturbs the frozen video encoder and can be ablated independently.

3.3. MGTA: Motion-Guided Temporal Adapter

MGTA models the video side. The frozen CLIP image encoder is frame-wise and static: it encodes each frame independently and has no cross-frame receptive field, so the onset, acceleration, direction change, and pose transition of an action are invisible to the backbone. MGTA therefore injects temporal-dynamics modeling into the frozen encoder while concentrating computation on the regions that actually move. The key observation is that an action is *localized and moving*: at any moment only a small part of the frame is action-critical (a hand, a club, a limb), and that part *changes position across frames*. A temporal module should therefore (i) *select* the discriminative regions rather than process the whole frame uniformly, (ii) *track* how those regions move over time, and (iii) do so in a way that leaves the frozen encoder untouched. MGTA realizes this with three components: *discriminative region selection* driven by a motion-aware score, *dual-branch temporal modeling* in which a dense Conv3d branch and a sparse asynchronous-attention branch complement each other, and *identity-preserving injection* so the module coincides with the baseline at initialization.

3.3.1. Discriminative region selection

Given the spatial-temporal patch tokens $F \in \mathbb{R}^{T \times N \times d}$ (T frames, N patches per frame), we score each patch by a normalized combination of three complementary cues:

$$s_{t,n} = \mathcal{Z}\Big(\alpha \, \hat{A}_{t,n} + (1 - \alpha) \, \hat{M}_{t,n} + \hat{\gamma}_{t,n}\Big), \tag{6}$$

where $\hat{A}_{t,n} = \mathcal{Z}(\text{Attn}(z^{\text{cls}}, f_{t,n}))$ is the *class-token attention*, $\hat{M}_{t,n} = \mathcal{Z}(\|f_{t,n} - f_{t-1,n}\|)$ is the *motion intensity*, $\hat{\gamma}_{t,n} = \mathcal{Z}(\gamma_{t,n})$ is a learnable saliency score, and $\mathcal{Z}$ denotes z-score normalization applied to each cue *before* combination. Each cue captures a distinct notion of "what matters" for an action: class-token attention locates the patches

that matter for the current class under the frozen representation; the motion-intensity term $\|f_{t,n} - f_{t-1,n}\|$ measures *where and how fast things change across frames*, which is precisely the temporal dynamics that a frame-wise image encoder cannot perceive; and the learnable saliency adapts the selection to the task at hand. Independent z-score normalization makes the three heterogeneous cues comparable in scale and prevents any single cue from dominating the score. The top-k patches with the highest scores are selected per frame as the discriminative region $\mathcal{S}$, intersected with a significance threshold so that a quiet frame cannot force a meaningless selection. Because the selection is recomputed *independently at each frame*, a moving part is re-located at its new spatial position in every frame rather than dropped after its first appearance; this frame-wise re-selection is what lets the module keep up with the motion of an action over time.

3.3.2. Dual-branch temporal modeling: dense locality and motion correspondence

Selection identifies *where* the action is, but the frozen encoder still has no way to read *how* those regions move. MGTA models the selected region with a dual-branch design whose costs are asymmetric by construction: a cheap *full-coverage* branch that guarantees dense local temporal modeling everywhere, and a *sparse* branch that explicitly predicts and exploits the motion of the action-critical tokens. The two branches answer complementary questions: "what changed locally?" and "where did it move to?"—and are fused at the patch level. The full-coverage branch is a depthwise Conv3d layer that slides over the *entire* spatial-temporal grid and captures dense local spatial-temporal patterns everywhere, providing a reliable lower bound on temporal modeling even for regions that the selection misses; because it is depthwise with a small kernel and shared weights, its cost stays linear in TN and independent of the selection. The sparse branch is the *Asynchronous Self-Attention* (ASA), which is applied *only to the selected top-k tokens* of each frame. Its core idea is an *implicit motion-correspondence*: rather than attending to a fixed neighborhood, each selected token predicts where the part it represents has moved to in the following frames, and then reads the features at that predicted location. Concretely, for each selected query q at its original position, a small offset network predicts a per-token 3D offset along the time, height, and width axes of the actual spatial grid,

$$\Delta_{t,n} = (\Delta_t, \Delta_h, \Delta_w) = \mathrm{offset_net}(q_{t,n}), \tag{7}$$

whose last layer is zero-initialized, so the offsets are zero at initialization and ASA starts from attention over the query's own local window. Because the offset is predicted from the query's *own* features, it encodes a *local* correspondence—"this part moves from here to there"—rather than a global uniform shift, which is what distinguishes it from shift-based temporal models [15]. Keys and values are then sampled from a $w \times w$ window ($w = 5$, i.e. $S = 25$ candidates) centered at the shifted position $(t+\Delta_t, h+\Delta_h, w+\Delta_w)$ using soft bilinear sampling along all three axes, which keeps the offsets fully differentiable:

$$\mathrm{ASA}(q) = \mathrm{softmax}\left(\frac{q\,K_{\text{samp}}^{\top}}{\sqrt{d}}\right) V_{\text{samp}}, \tag{8}$$

where $K_{\text{samp}}, V_{\text{samp}}$ are gathered at the offset-sampled candidate positions. The attention matrix is computed against *all* S sampled cross-frame candidates, so the query genuinely weighs how its motion-aligned neighborhood changes over time. This implicit motion-correspondence is what injects the dynamics that a frame-wise encoder misses: a query can *track a moving part to its new location in later frames* and read the feature that encodes how the part's pose or trajectory changed, effectively letting the sparse module act as a lightweight, learnable motion model. In contrast to full space-time attention that aggregates TN tokens per query, ASA attends to only S sampled candidates per selected query, so it concentrates the expensive cross-frame interaction on the action-critical regions that actually move.

3.3.3. Identity-preserving fusion and injection

The two branches answer different questions and should be combined adaptively. An *entry seed boost* first emphasizes the selected regions once on the raw tokens, so that the discriminative signal is highlighted before either branch runs. A patch-level gate, predicted from the temporal-mean statistics of both branch outputs and zero-initialized so that it outputs $\sigma = 0.5$ at initialization, lets the model allocate per patch between the dense local modeling and the motion-correspondence branch:

$$\delta_{t,n} = \sigma_{t,n}\,\mathrm{Conv3d}(f) + (1 - \sigma_{t,n})\,\mathrm{ASA}(f), \tag{9}$$

After fusion, the residual is written back to *every* patch position: the Conv3d branch modeled the whole grid, so unselected positions still receive the dense local temporal update, while the selected positions additionally get a confidence-proportional *soft-enhancement* so that action-critical parts are amplified without suppressing the background. Each frame's class token is then updated by a soft-weighted aggregation of the patch deltas, which provides a gradient bridge so that the discriminative signal reaches the class token that the downstream layers read. The entire module is *identity-preserving*: the offset network, the gate, and the enhancement scale are all zero-initialized, so at initialization the ASA reduces to local attention, the two branches are equally weighted, and the network coincides exactly with the plain frozen-CLIP baseline. This is what allows MGTA to be plugged into a frozen encoder with-

out any re-training of the backbone, and it lets the module be ablated independently. By keeping the dense branch full-coverage but cheap, and concentrating the expensive motion-correspondence attention on the selected top-k regions, MGTA both reduces redundancy and sharpens the discriminative signal, injecting temporal dynamics on top of a frozen encoder without modifying the CLIP backbone. Before the temporal mean, a lightweight frame-aggregation mixer applies a depthwise temporal convolution to the per-frame features; it is identity-start and adds negligible parameters, and its contribution is reported in the ablations.

3.4. BFEM: Bidirectional Feature Enhancement Module

BFEM enhances each modality with the other before classification. Because the text directions and the video class centers are induced by distinct modalities, supervision signals, and temporal aggregations, a video feature and its text prototype often disagree on the discriminative direction, so using one to refine the other can sharpen the representation for classification. BFEM therefore performs two directional *feature enhancement* operations under bounded, zero-initialized gates: the video feature is used to enhance the text direction, and the text prototype is used to enhance the video feature, so that the model starts from an identity mapping and gradually opens the gates during training. This warm-up avoids disturbing the already-converged adapters in the early stage, while still allowing a large correction once training stabilizes. The key property of the module is that every enhanced feature is a *convex combination* of the original feature and a modal counterpart: it stays on the segment between the two and can never be dragged outside the shared embedding ball, so the frozen representation remains the dominant signal.

3.4.1. Classification-driven soft enhancement

Given the L2-normalized video feature $v \in \mathbb{R}^d$ and text prototypes $Q \in \mathbb{R}^{C\times d}$ (each column a class direction), BFEM first computes a soft class assignment

$$p = \mathrm{softmax}\big(s\, vQ^\top\big)\,, \qquad \hat{p} = \mathrm{stopgrad}(p), \tag{10}$$

where s is a dedicated temperature ($s = 10$) rather than the shared CLIP logit-scale (100), which would saturate the softmax into one-hot and starve the enhancement path of gradients. The soft weights $\hat{p}$ are detached so that the loop $v \to p \to \hat{Q}, \hat{v} \to v$ is broken and the video's gradients come only from the structural enhancement terms, which stabilizes training. The probability $\hat{p}$ drives *both* directions below.

3.4.2. Video-to-text enhancement (V→T)

Each text direction is enhanced with the current video by an amount proportional to how likely the model considers that class:

$$\hat{Q}_b = \mathrm{normalize}(Q + g_t\, \hat{p}_b\, (v_b - Q))\,, \tag{11}$$

for every class c (index dropped), where g_t is a zero-initialized scalar gate bounded by $|g_t| \le 0.5$ via $\tanh$. Because row c rewrites as $(1 - g_t\hat{p}_{b,c})Q_c + g_t\hat{p}_{b,c}v_b$, the enhanced text is a convex combination of its original direction and the video; classes the model judges unlikely ($\hat{p}_{b,c} \approx 0$) keep their text unchanged, so the enhancement is confidence-conditioned rather than a uniform shift.

3.4.3. Text-to-video enhancement (T→V)

The video feature is enhanced with the text centroid $\hat{p}Q$ predicted by the model:

$$\hat{v}_b = \mathrm{normalize}(v_b + g_v\, (\hat{p}_b Q - v_b))\,, \tag{12}$$

where g_v is a *dynamic* gate driven by the class assignment itself, $g_v = \mathrm{max_gate} \cdot \tanh\big(\mathrm{MLP}(\hat{p}_b)\big)$ with a zero-initialized output layer, so it starts at zero and stays bounded. Equation (12) enhances v with the semantics of its predicted class, sharpening its discriminative direction without leaving the segment $[v, \hat{p}Q]$. Each direction optionally passes through a residual transform MLP ($D \to$ bottleneck $\to D$, output layer zero-initialized) that is identity at start, giving the module extra expressiveness once the gates wake up.

3.4.4. Training objective

The classification logit is computed as $s' \cos(\hat{v}_b, \hat{Q}_b)$ with the shared logit-scale $s' = \exp(\mathrm{logit_scale})$, and trained with a cross-entropy loss with label smoothing. Unlike attention-based reconstruction or projection schemes, BFEM needs *no auxiliary alignment loss and no prototypes*: the two enhancements are trained end-to-end by the classification objective alone, and the identity-start gates keep the first steps exactly equal to the frozen baseline. The only training-time scheduling is a gate warm-up that linearly ramps g_t, g_v from 0 over the first warm-up steps; at evaluation the gates are used at full strength.

3.5. Training details

The CLIP backbone is frozen. Only AMTF, MGTA, BFEM, the learnable branch's prompts, and the adapters are trainable. All three modules start from identity (zero-initialized gates and zero-initialized offsets), so the network behaves like the plain baseline at initialization. We train end-to-end with the cross-entropy objective and adopt the same optimization schedule as Vita-CLIP [27]. Algorithm 1 summarizes the overall training procedure. At each iteration, the frozen encoders map the sampled clip and the class prompts into tokens. AMTF builds the text prototypes, MGTA produces the video feature, and BFEM applies the bidirectional

Algorithm 1 MATE-CLIP training procedure

Input: frozen CLIP encoders, trainable $\Theta = \{$AMTF, MGTA, BFEM, prompts, adapter$\}$
Output: trained parameters Θ

initialize all gates to zero, offsets to zero
for epoch $= 1, \ldots, E$ **do**
  for each mini-batch of clips and labels **do**
    $F \leftarrow$ frozen image encoder (sampled T frames)
    $Q \leftarrow$ AMTF(frozen text encoder outputs) ▷ Sec. 3.2
    $\mathcal{S} \leftarrow$ discriminative region selection on F ▷ Eq. 6
    $v \leftarrow$ MGTA dual-branch temporal modeling on $\mathcal{S}$
    $\hat{p} \leftarrow$ stopgrad soft class assignment ▷ Eq. 10
    $\hat{v}, \hat{Q} \leftarrow$ BFEM bidirectional enhancement ▷ Eqs. 11,12
    compute $\mathcal{L} = \mathcal{L}_{\text{CE}}$
    update Θ by gradient descent
  end for
end for

enhancement before the cosine classifier computes the logits; no auxiliary alignment loss is needed. Because every module is identity-start, the first gradient step is taken from the converged baseline configuration, which stabilizes training of the gates and the offset network.

3.6. Computational cost

All three modules are lightweight and operate on the frozen CLIP features: AMTF and BFEM only add small linear layers, gates, and fusion operations, whose cost is negligible relative to the backbone. The dominant additional cost comes from MGTA, which we analyze next. A full space-time self-attention over all TN tokens costs $O(T^2N^2)$. MGTA instead keeps the dense branch cheap and makes only the attention sparse: it selects the top-k patches by score ($O(TN)$ scoring plus a top-k selection) and applies ASA to the selected tokens only, so the attention cost reduces to $O(kS)$ per frame with $k \ll TN$ and $S \ll TN$, while the depthwise Conv3d branch covers the full grid at $O(TN)$ with a small constant. The offset network is a tiny MLP shared across tokens, adding only $O(TN)$ parameters. Consequently, MGTA injects temporal-dynamics modeling at a fraction of the cost of full space-time attention, while the frozen backbone ensures the total trainable parameters remain small.

Table 2 compares the asymptotic cost of the main video-modeling choices. With $T = 8$ frames and $N = 196$ patches per frame (ViT-B/16 at 224^2), a full space-time attention touches 1,568 tokens per query and costs $\sim 2.5 \times 10^6$ attention FLOPs per query, whereas MGTA restricts the ASA to the top-$k = 12$ patches per frame with $S = 25$ sampled candidates per query, reducing the attention cost per query by more than an order of magnitude. The selection itself is $O(TN)$ and is shared across all queries, and the depthwise Conv3d branch covers the full grid at a linear $O(TN)$ cost independent of the selection, so the total additional cost scales with the sparse attention term $O(kS)$ plus a small linear term, rather than with the quadratic $O(T^2N^2)$ of full space-time attention. AMTF and BFEM add a few linear projections and fusion/enhancement operations whose cost is negligible compared with the backbone, and the total number of trainable parameters added by MATE-CLIP remains a small fraction of the frozen CLIP ViT-B/16 backbone, as we report in the experiments.

Method	Attention FLOPs	Param.
Full space-time attention	$O(T^2N^2)$	high
Uniform token aggregation	$O(TN)$	low
MGTA (ours)	$O(TN + kS)$	low

Table 2. Asymptotic cost of video-modeling strategies. $k \ll TN$ denotes selected patches.

4. Experiments

4.1. Datasets

We evaluate on two standard benchmarks to assess both accuracy and cross-dataset generalization of the frozen-CLIP adaptation. **UCF101** [20] contains 13,320 realistic action videos from 101 categories collected from YouTube; we follow the standard split 1 (9,537 training and 3,783 testing clips), and its categories span human–object interaction, body-motion only, human–human interaction, playing musical instruments, and sports, with substantial intra-class variation in viewpoint, scale, and background. **HMDB-51** [10] contains 6,766 clips from 51 action categories (mostly full-body actions), each with a limited number of samples, which makes it a harder and more motion-centric benchmark for evaluating temporal modeling and generalization of the frozen-CLIP adaptation. Together, the two datasets let us verify that the gains of MATE-CLIP are not specific to a single benchmark.

4.2. Implementation details

We build on the CLIP ViT-B/16 backbone [19] and keep both encoders frozen throughout training. Videos are sampled into 8 frames at a stride chosen so that the clip covers the action, and evaluation uses 3 spatial and 3 temporal views. The frozen backbone is augmented with the RAP-style correlated adapter as the base video model [2]. MGTA is applied on adapter layers $\{3, 5, 7, 9\}$ with top-k patches $k = 12$, 8 asynchronous attention heads, a space-time offset bound of 1.0 (temporal) and 4.0 (spatial) patches, and a window size of 5 frames. BFEM uses a dedicated classifica-

Hyper-parameter	Value
Backbone	CLIP ViT-B/16 (frozen)
Input frames T	8
Patch size	16×16
MGTA layers	$\{3, 5, 7, 9\}$
Selected patches k	12
ASA heads	8
Offset bound (temp., spat.)	1.0, 4.0
Temporal window	5
BFEM temp. s	10
BFEM max gate	0.5
BFEM gate warm-up	100 steps
Epochs / warm-up	60 / 5
Base learning rate	10^{-4}
Label smoothing	0.1
Effective batch size	64
Test views	3×3

Table 3. Hyper-parameters of MATE-CLIP.

tion temperature of $s = 10$, enhancement gates bounded by $\max_gate = 0.5$ and warmed up over the first 100 steps, and optional per-direction residual transforms with a bottleneck of 128; no alignment loss is used. We train for 60 epochs with 5 warm-up epochs, a base learning rate of 10^{-4} for the modules (with separate learning rates for prompts and adapters), label smoothing of 0.1, and gradient accumulation that yields an effective batch size of 64. The backbone is loaded from the official CLIP weights and is not updated. Table 3 lists the full hyper-parameter configuration.

4.3. Evaluation protocol and baselines

We report top-1 accuracy (%). Our primary baselines are (i) the frozen CLIP baseline with the text `[EOT]` representation and uniform frame pooling, (ii) Vita-CLIP [27], a strong frozen-CLIP method with multimodal prompting, and (iii) RAP [2], the sparse-and-correlated adapter that serves as the direct sparse baseline for MGTA. We also compare with representative video transformers trained from scratch on the same benchmark [1, 5, 16] to situate frozen-CLIP methods in the literature. All methods are evaluated under the same protocol: 8-frame clips, the same test-time augmentation, and top-1 accuracy reported on the standard split 1. For the frozen CLIP and Vita-CLIP baselines we reuse the official hyper-parameters, and for RAP we use the configuration that matches our base model so that the gains of MGTA are attributable to the module itself rather than to a stronger base.

4.4. Main results

Table 4 reports the main comparison. On both UCF101 and HMDB-51, the frozen CLIP baseline with the `[EOT]` representation and uniform pooling already performs reasonably, and the RAP sparse adapter improves it further, confirming that focusing on informative patches helps even when scenes and background are shared across many categories. MATE-CLIP is expected to further improve over RAP by adding the temporal-dynamics modeling of MGTA, the action-biased text prototypes of AMTF, and the bidirectional enhancement of BFEM. The full MATE-CLIP model combines all three modules. Numbers marked **[TODO:]** are pending training completion and will be filled from the training logs.

Method	UCF101	HMDB-51
TimeSformer [1]	95.6	–
Video Swin-T [16]	96.4	–
Frozen CLIP baseline	–	–
Vita-CLIP [27]	–	–
RAP baseline [2]	93.84	–
MGTA	**[TODO:]**	**[TODO:]**
+ AMTF	**[TODO:]**	**[TODO:]**
MATE-CLIP (full)	**[TODO:]**	**[TODO:]**

Table 4. Top-1 accuracy (%) on UCF101 and HMDB-51. Numbers marked **[TODO:]** are pending training completion and will be filled from the training logs.

4.5. Efficiency analysis

Beyond accuracy, we measure the computational cost introduced by each module, because a lightweight design is a core requirement of frozen-CLIP adaptation. Table 5 reports the number of additional trainable parameters and the throughput measured on a single GPU (to be filled). As analyzed in Section 3.6, AMTF and BFEM add only a few linear layers and enhancement operations, and their parameter growth is negligible. MGTA adds the offset network and two lightweight branches but keeps the expensive attention sparse: the motion-correspondence attention runs only on the selected top-k patches, while the depthwise Conv3d branch stays full-coverage yet linear in the token count, so the overall cost is proportional to k plus a small $O(TN)$ term rather than to the quadratic cost of full space-time attention. Compared with full space-time attention, MGTA reduces the attention cost by more than an order of magnitude while injecting the temporal dynamics that a frame-wise encoder lacks.

4.6. Ablation studies

Our ablations address the modules in turn, and each module is ablated along the dimension where its design choice matters most. For the text side, we ablate the static and learnable branches (each alone vs. their fusion), and within AMTF we compare the `[EOT]`-only view with adding the masked-mean and the masked-attention views; we report the learned branch-level fusion weights to show when

Method	#Params	Throughput
Frozen CLIP baseline	–	–
RAP baseline	–	–
+ MGTA	–	–
+ AMTF	–	–
+ BFEM	–	–
MATE-CLIP (full)	–	–

Table 5. Parameter and throughput comparison (to be filled after training).

MGTA variant	UCF101	HMDB-51
Uniform selection	–	–
Class-token-only selection	–	–
+ motion intensity (ours)	–	–
$k = 6$	–	–
$k = 12$ (ours)	–	–
$k = 24$	–	–
Conv3d only	–	–
ASA only	–	–
MGTA (both branches)	–	–

Table 6. MGTA ablation design on UCF101 and HMDB-51. Results are pending training completion.

the static semantic bank versus the dynamic action-biased branch is preferred. For MGTA, we vary the number of selected patches k, ablate the two branches (Conv3d vs. asynchronous motion-correspondence attention) independently, and compare motion-aware selection (with the motion-intensity cue) against uniform selection and class-token-only selection; we additionally ablate the learned offsets against fixed zero offsets, which isolates the contribution of the implicit motion correspondence itself. For BFEM, we ablate the two enhancement directions (video-to-text vs. text-to-video), the dynamic versus scalar gate, and the zero-initialized gate warm-up schedule, to verify that the bidirectional design and the warm-up are both necessary. Finally, we combine the modules one at a time to verify that the gains are complementary and orthogonal. Table 6 shows the MGTA ablation design. The remaining ablations follow the same structure (tables to be filled after training). The key expected findings are: (i) motion-intensity guidance contributes more for categories whose classes share similar scenes and are separated mainly by body motion, because the motion cue disambiguates classes that appearance cannot; (ii) the asynchronous attention branch matters most when actions involve large spatial displacement, such as twists and somersaults; and (iii) the zero-initialized gates are essential, since removing them would make the injection start from a non-identity configuration and destabilize training.

4.7. Robustness and generalization

To understand when MATE-CLIP helps and whether it generalizes, we analyze sensitivity to the number of input frames and the sparsity level k, and evaluate consistently on both UCF101 and HMDB-51. Using fewer frames (e.g., 4) stresses the temporal modeling because motion is undersampled; using more frames (e.g., 16) increases redundancy and the risk of diluting the discriminative signal. We expect MGTA to remain accurate at 4 frames because the motion-intensity cue still identifies the moving parts, and to saturate at 16 frames with little extra cost, since the attention is sparse. The sparsity level k trades accuracy for cost: too small a k drops the informative patches, while too large a k re-introduces the dilution that sparsity is meant to remove. Crucially, because AMTF and MGTA adapt the frozen CLIP without task-specific re-training of the backbone, the same model transferred across UCF101 and HMDB-51 preserves its gains, evidencing that the adaptation generalizes rather than overfitting a single benchmark. These experiments, together with the qualitative analysis below, will be reported after training completes.

4.8. Qualitative analysis

We provide three qualitative views (to be filled after training): (i) region-selection visualization, showing that MGTA selects the moving parts (e.g., an athlete's limbs during a gymnastics routine) across frames rather than static background; (ii) text-prototype visualization, illustrating that AMTF biases the text representation toward the action-related cues of the learnable branch while staying anchored to the static semantic bank; and (iii) an alignment t-SNE of text directions and video class centers, demonstrating that BFEM reduces the cross-modal shift before classification.

5. Conclusion

We presented MATE-CLIP, a parameter-efficient framework for multimodal adaptation and temporal enhancement that adapts a frozen CLIP backbone to video action recognition. We identified three complementary semantic gaps that jointly limit frozen CLIP video recognition: text-side information loss, video-side temporal-dynamics deficiency with discriminative dilution, and cross-modal directional misalignment. We addressed them with three plug-and-play modules, each targeting one stage of the information pipeline: AMTF performs multi-strategy text fusion on top of a static semantic branch so that the text prototypes go beyond the fixed generic template and capture the true action semantics, MGTA injects temporal-dynamics modeling through motion-guided region selection and implicit motion-correspondence attention that tracks where the action-critical regions move across frames, and BFEM bidirectionally enhances the video and text features with each

other before classification. All modules operate on a frozen CLIP backbone with zero-initialized or identity-start gates, so the network coincides with the plain baseline at initialization and the CLIP backbone is never modified. Experiments on UCF101 and HMDB-51 show consistent improvements over strong CLIP-based baselines and confirm that the adaptation generalizes across datasets, and ablations verify the orthogonality and complementarity of the three modules. We believe MATE-CLIP provides a simple yet effective recipe for adapting frozen CLIP models to video action recognition.

A natural direction for future work is to extend MATE-CLIP to the open-vocabulary and zero-shot settings, where the text anchors must generalize beyond the training classes and the alignment should not require class-specific prototypes. Another direction is to integrate MGTA with stronger base models (e.g., VideoMamba [14] or Video Swin [16]) and to study the interaction between sparse attention and long-video temporal structure. Finally, the identity-start property of our modules suggests that they can be composed with other parameter-efficient adapters [7, 18] without retraining the backbone, which we leave as future work.

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